# Operating System Assignment 2

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Title: Execution of Advanced Linux commands.

Aim: To study and execute advanced Linux commands for file operations, process management, and user management using the Ubuntu Linux operating system.

Objective:

- To understand the Linux command-line interface (CLI).
- To perform file and directory operations using Linux commands.
- To manage running processes using process management commands.
- To create, modify, and manage user accounts using user management commands.
- To gain practical experience in using the Linux operating system.

Relative Theory:

Linux is an open-source, multi-user operating system that uses a Command Line Interface (CLI) to perform tasks. Linux commands help users manage files, processes, and user accounts efficiently. This practical provides hands-on experience with commonly used Linux commands.

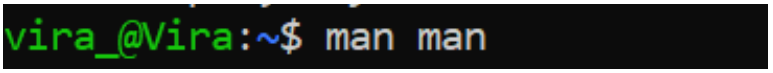
Execution:

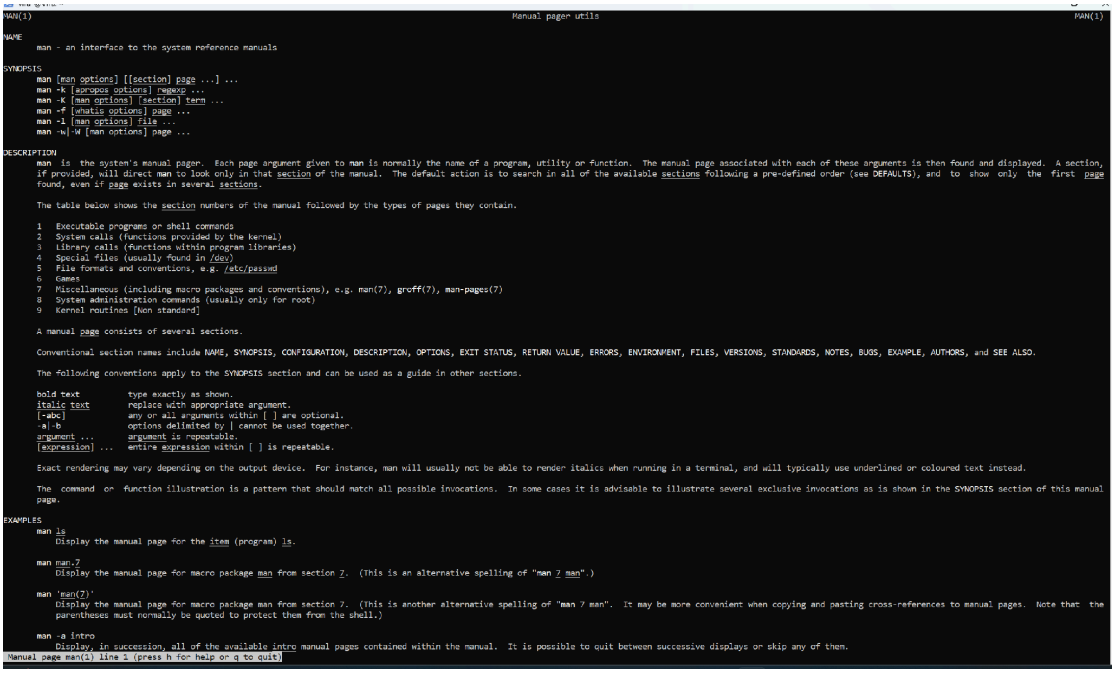
1. clear




2. man





```
MAN(1)                                   Manual pager utils                                   MAN(1)

NAME
       man - an interface to the system reference manuals

SYNOPSIS
       man [man options] [[section] page ...] ...
       man -k [apropos options] regexp ...
       man -K [man options] [section] term ...
       man -f [whatis options] page ...
       man -l [man options] file ...
       man -w|-W [man options] page ...

DESCRIPTION
       man  is  the system's manual pager.  Each page argument given to man is normally the name of a program, utility or function.  The manual page associated with each of these arguments is then found and displayed.  A section,
       if provided, will direct man to look only in that section of the manual.  The default action is to search in all of the available sections following a pre-defined order (see DEFAULTS), and  to  show  only  the  first  page
       found, even if page exists in several sections.

       The table below shows the section numbers of the manual followed by the types of pages they contain.

       1   Executable programs or shell commands
       2   System calls (functions provided by the kernel)
       3   Library calls (functions within program libraries)
       4   Special files (usually found in /dev)
       5   File formats and conventions, e.g. /etc/passwd
       6   Games
       7   Miscellaneous (including macro packages and conventions), e.g. man(7), groff(7), man-pages(7)
       8   System administration commands (usually only for root)
       9   Kernel routines [Non standard]

       A manual page consists of several sections.

       Conventional section names include NAME, SYNOPSIS, CONFIGURATION, DESCRIPTION, OPTIONS, EXIT STATUS, RETURN VALUE, ERRORS, ENVIRONMENT, FILES, VERSIONS, STANDARDS, NOTES, BUGS, EXAMPLE, AUTHORS, and SEE ALSO.

       The following conventions apply to the SYNOPSIS section and can be used as a guide in other sections.

       bold text          type exactly as shown.
       italic text        replace with appropriate argument.
       [-abc]             any or all arguments within [ ] are optional.
       -a|-b              options delimited by | cannot be used together.
       argument ...       argument is repeatable.
       [expression] ...   entire expression within [ ] is repeatable.

       Exact rendering may vary depending on the output device.  For instance, man will usually not be able to render italics when running in a terminal, and will typically use underlined or coloured text instead.

       The  command  or  function illustration is a pattern that should match all possible invocations.  In some cases it is advisable to illustrate several exclusive invocations as is shown in the SYNOPSIS section of this manual
       page.

EXAMPLES
       man ls
           Display the manual page for the item (program) ls.

       man man.7
           Display the manual page for macro package man from section 7.  (This is an alternative spelling of "man 7 man".)

       man 'man(7)'
           Display the manual page for macro package man from section 7.  (This is another alternative spelling of "man 7 man".  It may be more convenient when copying and pasting cross-references to manual pages.  Note that  the
           parentheses must normally be quoted to protect them from the shell.)

       man -a intro
           Display, in succession, all of the available intro manual pages contained within the manual.  It is possible to quit between successive displays or skip any of them.
Manual page man(1) line 1 (press h for help or q to quit)
```

3. ps aux

```
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24436 15388 ?        Ss   11:20   0:00 /sbin/init
root           2  0.0  0.0   3172  2208 hvc0     Sl+  11:20   0:00 /init
root           6  0.0  0.0   3204  2140 hvc0     Sl+  11:20   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          56  0.0  0.2  66768 20256 ?        S<s  11:20   0:00 /usr/lib/systemd/systemd-journald
systemd+      92  0.0  0.1  22524 14304 ?        Ss   11:20   0:00 /usr/lib/systemd/systemd-resolved
root         103  0.0  0.1  35252 12128 ?        Ss   11:20   0:02 /usr/lib/systemd/systemd-udevd
root         183  0.0  0.0   2880  1912 ?        Ss   11:20   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         184  0.0  0.0   4464  3064 ?        Ss   11:20   0:00 /usr/sbin/cron -f -P
message+     185  0.0  0.0   8816  5280 ?        Ss   11:20   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         189  0.0  0.3  44076 29468 ?        Ss   11:20   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         199  0.0  0.1  18476  9420 ?        Ss   11:20   0:00 /usr/lib/systemd/systemd-logind
syslog       225  0.0  0.0 220532  5604 ?        Ssl  11:20   0:00 /usr/sbin/rsyslogd -n -iNONE
root         232  0.0  0.0   5484  2680 tty1     Ss+  11:20   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d - linux
root         246  0.0  0.4 123440 32272 ?        Ssl  11:20   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
_chrony      256  0.0  0.1  24508 11636 ?        S    11:20   0:00 /usr/sbin/chronyd -n -F 1 -x
_chrony      263  0.0  0.0  12076  2424 ?        S    11:20   0:00 /usr/sbin/chronyd -n -F 1 -x
root         441  0.0  0.0   3188  1116 ?        Ss   11:20   0:00 /init
root         442  0.0  0.0   3188  1248 ?        S    11:20   0:00 /init
vira_        443  0.0  0.0   6840  6112 pts/0    Ss   11:20   0:00 /bin/bash
vira_        680  0.0  0.1  22728 13260 ?        Ss   11:22   0:00 /usr/lib/systemd/systemd --user
vira_        682  0.0  0.0  22896  4116 ?        S    11:22   0:00 (sd-pam)
root         794  0.0  0.0   8632  5504 ?        Ss   11:22   0:00 login -- vira_
vira_        839  0.0  0.0   6128  5356 pts/1    Ss+  11:22   0:00 -bash
root        2331  0.0  0.0  35256  6464 ?        S    12:10   0:00 (udev-worker)
root        2332  0.0  0.0  35256  6464 ?        S    12:10   0:00 (udev-worker)
vira_       2333  0.0  0.0   7152  4332 pts/0    R+   12:10   0:00 ps aux
```

4. top

```
top - 12:10:17 up 49 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7792.1 total,   7104.4 free,    588.1 used,    252.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7203.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24436  15388  11332 S   0.0   0.2   0:00.63 systemd
      2 root      20   0    3172   2208   2072 S   0.0   0.0   0:00.04 init-systemd(Ub
      6 root      20   0    3204   2140   2008 S   0.0   0.0   0:00.00 init
     56 root      19  -1   66768  20256  18956 S   0.0   0.3   0:00.15 systemd-journal
     92 systemd+  20   0   22524  14304  11696 S   0.0   0.2   0:00.03 systemd-resolve
    103 root      20   0   35252  12128   9012 S   0.0   0.2   0:02.78 systemd-udevd
    183 root      20   0    2880   1912   1784 S   0.0   0.0   0:00.03 chronyd-starter
    184 root      20   0    4464   3064   2816 S   0.0   0.0   0:00.00 cron
    185 message+  20   0    8816   5280   4588 S   0.0   0.1   0:00.08 dbus-daemon
    189 root      20   0   44076  29468  14340 S   0.0   0.4   0:00.14 networkd-dispat
    199 root      20   0   18476   9420   8108 S   0.0   0.1   0:00.09 systemd-logind
    225 syslog    20   0  220532   5604   4508 S   0.0   0.1   0:00.07 rsyslogd
    232 root      20   0    5484   2680   2492 S   0.0   0.0   0:00.00 agetty
    246 root      20   0  123440  32272  17672 S   0.0   0.4   0:00.12 unattended-upgr
    256 _chrony   20   0   24508  11636   6884 S   0.0   0.1   0:00.09 chronyd
    263 _chrony   20   0   12076   2424   1788 S   0.0   0.0   0:00.00 chronyd
    441 root      20   0    3188   1116    980 S   0.0   0.0   0:00.00 SessionLeader
    442 root      20   0    3188   1248   1108 S   0.0   0.0   0:00.16 Relay(443)
    443 vira_     20   0    6840   6112   3868 S   0.0   0.1   0:00.16 bash
    680 vira_     20   0   22728  13260  10812 S   0.0   0.2   0:00.06 systemd
    682 vira_     20   0   22896   4116   2116 S   0.0   0.1   0:00.00 (sd-pam)
    794 root      20   0    8632   5504   4668 S   0.0   0.1   0:00.00 login
    839 vira_     20   0    6128   5356   3804 S   0.0   0.1   0:00.01 bash
   2336 vira_     20   0    8208   6060   3912 R   0.0   0.1   0:00.00 top
   2337 root      20   0   35256   6464   3316 S   0.0   0.1   0:00.00 (udev-worker)
   2338 root      20   0   35256   6464   3316 S   0.0   0.1   0:00.00 (udev-worker)
```

5. head

```
~$ echo "Line 1" > sample.txt
~$ echo "Line 2" >> sample.txt
~$ echo "Line 3" >> sample.txt
~$ echo "Line 4" >> sample.txt
~$ echo "Line 5" >> sample.txt
~$ echo "Line 6" >> sample.txt
~$ echo "Line 7" >> sample.txt
~$ echo "Line 8" >> sample.txt
~$ echo "Line 9" >> sample.txt
~$ echo "Line 10" >> sample.txt
~$ echo "Line 11" >> sample.txt
~$ head sample.txt
```

```
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
```

6. tail

```
~$ tail sample.txt
```

```
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
```